WetSpass-M Modern

User Guide



Version 2.4.1

M. G. Gurmu (Senior Hydrogeologist)
mggurmu@gmail.com

October 2026 (Addis Ababa)



Preface

This document is the user guide for WetSpass-M Modern, a modernization of the original Wetspass-M 2016 model developed by K. Abdollahi, I. Bashir, and O. Batelaan at the Vrije Universiteit Brussel. WetSpass-M is one of the most widely used spatially distributed water-balance models for estimating groundwater recharge at catchment and basin scale, particularly in data-scarce regions where simpler lumped models are insufficient and full physically based models are impractical. WetSpass-M Modern preserves the original model's physics, equations, and lookup-table structure; what it changes is the way the soil column is treated over time, the way the annual mass balance is defined, and the way the model is delivered to the user.

The improvements documented in this manual are:

- **A dynamic soil-water bucket** replacing the original quasi-steady-state assumption. Soil moisture is carried as an explicit state variable from one month to the next, so the seasonal redistribution of water through the root zone is preserved. This matters most in seasonally concentrated rainfall regimes — monsoonal, Mediterranean, semi-arid, and arid climates — where the original assumption of monthly reset can distort the seasonal timing of recharge and the dry-season supply for evapotranspiration.
- **An ET cap consistent with mass balance.** Soil-derived evapotranspiration is capped at the water physically available after runoff (P − SR), not at rainfall alone. This prevents the model from producing negative recharge on high-runoff cells and ensures the catchment-scale balance closes.
- **A separate diagnostic for groundwater-fed ET.** Transpiration and evaporation drawn from a shallow water table are computed explicitly and reported as `Cell_gw_discharge`, outside the precipitation balance, instead of being folded back into AET where they would double-count water in the closure test.
- **Integration with QGIS via GeoTIFF I/O.** Inputs and outputs are GeoTIFF rasters. The plugin runs inside QGIS, uses QGIS's own raster and vector libraries, and writes outputs that can be inspected, symbolised, and analysed directly in the QGIS map canvas. Legacy ASCII `.asc` grids are still accepted for backward compatibility.
- **Robust checkpointing and resume for long runs.** The workflow saves its state after every completed month and after every completed spin-up cycle. A power loss, an external drive disconnection, or an unexpected shutdown no longer means starting over — the next time the plugin is opened, the Run tab offers a Resume run button that continues from the last completed month. Only one month of computation is ever at risk.
- **A Focused Recharge module** for streambed infiltration along ephemeral and intermittent stream networks, with a user-selectable combine method for diffusive and focused recharge and a climate-zone-based physical cap derived from mean annual precipitation.
- **A Crop outputs to boundary tool** that replaces the failing QGIS `gdal:rasterize` "Mask rasters" workflow. It presents a file-selection dialog so you choose exactly which rasters to crop, auto-reprojects the boundary vector to each raster's CRS, retains edge pixels, and writes cropped

copies to a separate folder without modifying the originals.

The manual assumes you are familiar with QGIS, raster GIS concepts, and basic hydrology (rainfall, runoff, evapotranspiration, recharge). If you are new to any of those, work through the Quick Start in Chapter 3 first — it walks you from a clean QGIS install to a valid annual water balance in about five minutes using the sample data. Consult the Physics and equations chapter only when you need to understand a specific flux or make a calibration decision. The Troubleshooting chapter at the end collects the errors most commonly reported by users, together with their causes and fixes.

Table of Contents

Preface i
Table of Contents iv
1. Introduction and scope 1
1.1 What WetSpass-M Modern does 1
1.2 Who should use it 1
1.3 Model type 1
1.4 What changed from the 2016 original 2
1.5 What is new in 2.4.1 2
2. Installation 4
2.1 Requirements 4
2.2 Installing the plugin 4
2.3 Verifying dependencies 4
2.4 Shipping the user guide 4
3. Quick start in five minutes 5
4. Input data requirements 6
4.1 Standard folder layout 6
4.2 Grid alignment 6
4.3 Raster formats 7
4.4 Units 7
4.5 Monthly file naming 7
4.6 Lookup tables 7
4.7 Optional inputs 8
5. Tab-by-tab reference 9
5.1 Inputs tab 9
5.2 Check Inputs tab 9
5.3 Settings tab 9
5.4 Parameters tab 10
5.5 Output Period tab 10
5.6 Run tab 10
5.7 Progress tab 11
5.8 Results tab 11
5.9 Water Balance tab 11
5.10 Focused Recharge tab 11
5.11 Crop outputs to boundary 11

5.12 About tab 11
5.13 Help tab 11
5.14 Calibration tab 12
6. Physics and equations 13
6.1 Water balance at a cell, per month 13
6.2 Interception 13
6.3 Surface runoff 13
6.4 Actual evapotranspiration 14
6.5 Recharge 14
6.6 Snowmelt 14
7. Spin-up and convergence 15
7.1 Why spin-up is needed 15
7.2 Recommended spin-up counts 15
7.3 Automatic convergence check 15
7.4 Convergence report 16
7.5 Manual verification 16
8. Outputs and interpretation 17
8.1 Output folder structure 17
8.2 Output variables 17
8.3 Reading the annual rasters 18
8.4 Simulated.tbl 18
9. The Focused Recharge module 19
9.1 What it computes 19
9.2 Parameters 19
9.3 Climate regime 19
9.4 Combine methods 20
9.5 Physical cap 20
9.6 Workflow 21
10. Crop outputs to boundary 22
10.1 What it does 22
10.2 Why it exists 22
10.3 Selection dialog 22
10.4 Where the output goes 22
10.5 Behaviour 22
10.6 Recommended workflow 23
11. The water balance panel 24
11.1 Reading the panel 24
11.2 What ΔS means 24

11.3 Focused recharge integration 24
12. Compatibility mode 25
12.1 What it does 25
12.2 When to use it 25
13. Calibration and validation 26
13.1 What the two words mean in this context 26
13.2 What the Calibration tab does 26
13.3 The metrics 26
13.4 What to calibrate against 27
13.5 Calibration workflow 27
13.6 Target values 28
13.7 Validation protocol 28
13.8 Common calibration pitfalls 29
14. Troubleshooting 30
14.1 The Check Inputs tab shows red rows 30
14.2 "Input rasters are not on the same grid as the DEM" 30
14.3 Recharge raster is all zeros 30
14.4 Closure error is large in the Water Balance panel 30
14.5 Checkpoint errors on Resume 30
14.6 Model appears stuck on a spin-up cycle 30
14.7 Power loss or drive disconnection during a run 31
14.8 "Output device unavailable" warning 31
14.9 Duplicate "[Check Inputs]" log lines 31
14.10 Crop tool: "geopandas is not installed" 31
14.11 Crop tool: "Invalid path — Output directory does not exist" 31
14.12 Crop tool: some files fail to crop 31
14.13 Focused recharge: "operands could not be broadcast together" 31
14.14 Focused recharge: outliers in Recharge_focused_annual.tif 32
14.15 Combine: "Duplicate argument -ot" 32
14.16 Which combine method should I use 32
14.17 The capped total looks too low 32
15. Advanced Optional Features 33
15.1 Auto-Cropping to Boundary 33
15.2 Groundwater Depth (GWD) Capping (UNEP 5-Zone Climate) 33
15.3 Spin-Up Convergence Criteria 34
16. River-Aquifer Conductance & Focused Recharge (Experimental) 35
16.1 Introduction & Novel Contributions 35

16.2 Governing Equations & Units 35
17. References and acknowledgements 46
17.1 Original model 46
17.2 Bucket formulation 46
17.3 Evapotranspiration 46
17.4 Conductance Related 46
17.5 Acknowledgements 47
Appendix A — Glossary 48
Appendix B — File listing 50
Appendix C - Detailed Breakdown of Each Coefficient (River Bed Conductance) 51
Appendix D - Indicators for Calibration (What to look for) 53

1. Introduction and scope

1.1 What WetSpass-M Modern does

WetSpass-M Modern computes, on a regular grid and at monthly time steps:

- Interception by vegetation.
- Surface runoff, split by land-cover fraction.
- Actual evapotranspiration (transpiration + soil evaporation + open-water evaporation + impervious-surface evaporation).
- Diffuse recharge to the saturated zone.
- (Optionally) focused recharge along ephemeral stream networks.

The output is a set of GeoTIFF rasters at monthly, seasonal, and annual resolution, together with a text summary file (`Simulated.tbl`), a convergence report (`spinup_convergence.csv`), and a run report (`RunSummary.txt` plus `_run_metadata.json`).

1.2 Who should use it

The model is designed for:

- Regional groundwater recharge estimation.
- Water-balance studies at catchment to basin scale.
- Long-term average annual and seasonal water budgets.
- Climate-change impact assessment (via substituting future climate inputs).
- Land-use change scenario analysis (via substituting land-use maps).

It is not designed for event-scale flood modelling, sub-monthly routing, or transient groundwater simulation. For those, use a physically-based rainfall-runoff model coupled to a groundwater model.

1.3 Model type

Two operating modes are available:

- **Dynamic bucket mode (default).** Soil moisture is an explicit state carried from month to month. Mass balance: $P = AET + SR + R + \Delta S$. At the annual scale, after spin-up, $\Delta S \approx 0$.
- **Wetspass-M 2016 compatibility mode.** Soil moisture is reset to zero every month ($\Delta S \equiv 0$). This reproduces the original 2016 physics exactly and is intended for validation against legacy results.

1.4 What changed from the 2016 original

Aspect	Wetspass-M 2016	WetSpass-M Modern
Soil moisture	Reset every month (ΔS = 0)	Dynamic bucket, carried forward
ET cap	At rainfall	At (rainfall − runoff)
Groundwater-fed ET	Folded into AET	Reported separately as Cell_gw_discharge
Time integration	Monthly, one pass	Monthly, spin-up + reported pass
I/O	ASCII grids	GeoTIFF with ASCII fallback
Aggregation	Simple sum	Nodata-preserving sum
Environment	Standalone GUI	QGIS plugin
Combine of diffusive + focused	N/A	User-selectable, with climate cap
Boundary cropping	Manual in GIS	Built-in, with file-selection dialog

1.5 What is new in 2.4.1

This version, **v2.4.1**, represents a major architectural leap forward, introducing advanced MODFLOW integration and robust workflow management. It adds several substantial capabilities on top of the previous release:

- ✓ **River-Aquifer Conductance Module:** A completely new tab dedicated to MODFLOW 6 integration. It uses a novel Field-Anchored Geomorphic Scaling methodology to propagate outlet field measurements (Conductance, Depth, Stage) upstream using the Outlet Expansion Index (OEI) and Strahler order. It includes optional Stream Typology and spatial Geology raster multipliers.
- ✓ **MODFLOW 6 RIV Package Export:** Automatically generates a `MODFLOW_Stream_Table.csv` containing Layer, Row, Column, absolute Stage elevation, Conductance (m^2/day), and River Bottom elevation—ready for direct import into ModelMuse.
- ✓ **Multi-Flow Direction (MFD) Engine:** The stream derivation engine was upgraded from single-flow D8 to MFD to permanently solve the "straight-line" stream artifacts that occur in flat, arid plains.
- ✓ **Smart UNEP Capping & Cache Management:** The UNEP 5-Zone Climate Capping now features a smart cache. If a user toggles the capping off, or changes the manual override, the system automatically clears the old `effective_gwdepth` folder and recalculates on the fly.
- ✓ **Instant "Skip to Final Cycle":** The skip button now instantly aborts the current month and block, immediately jumping to the final output cycle rather than waiting for the 12-month spin-up to finish.
- ✓ **Active Workflow Control:** The "Run" button remains active even when a paused checkpoint exists, prompting the user to either resume or start a fresh run.
- ✓ **Intelligent Auto-Cropping:** The workflow now automatically crops all input rasters to the project boundary shapefile before processing, preventing RAM crashes on country-scale grids. The process is resumable if interrupted.
- ✓ **Robust Grid Alignment for Combine:** The "Combine with Diffusive Recharge" tool now automatically resamples the focused recharge raster onto the diffusive grid before addition, preventing `broadcast` shape errors.

Bug Fixes & Stability:

- ✓ Resolved the infinite mass-balance errors (-48 mm) in humid lake catchments (like Lake Chamo) by allowing users to disable the UNEP capping and use raw groundwater depths.
- ✓ Fixed the Streambed Depth calculation, preventing the depth from exploding to 43+ meters due to a GRASS Strahler ID extraction bug.
- ✓ Resolved a critical grid shape mismatch error when combining diffusive and focused recharge.
- ✓ Fixed the CSV output units to ensure Focused Recharge is correctly reported in mm/year*mm/year*.

The previous version, 2.2.1, added three substantial capabilities on top of the preceding releases:

- A **Crop outputs to boundary** tool that replaces the failing QGIS `gdal:rasterize` workflow with a robust `rasterio`-based implementation and a file-selection dialog.
- A **climate-zone-based physical cap** on total recharge, derived automatically from mean annual precipitation and the standard reference table.
- A **user-selectable combine method** for diffusive and focused recharge — addition, replace-by-max, or focused-over-diffusive — with a Preview button that reports what a given cap would do before any file is written.

The release also fixes three defects in the focused-recharge engine that had been producing physically impossible recharge values and blocking runs with broadcast and rasterization errors. Diffusive recharge remains untouched: it is produced by the same workflow, with the same equations and the same spin-up logic as version 2.2.1.

2. Installation

2.1 Requirements

- QGIS 3.28 or newer (tested through 3.44 and listed for QGIS 4.x via `qgisMaximumVersion=4.99`).
- Python 3.9 or newer (comes bundled with QGIS).
- Python packages: `numpy`, `rasterio`, `geopandas`. The first two are commonly available in QGIS's Python environment. `geopandas` may need to be installed separately for the crop tool.

2.2 Installing the plugin

Locate your QGIS plugin directory:

- **Windows:** `%APPDATA%\QGIS\QGIS3\profiles\default\python\plugins\`
- **Linux:** `~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/`
- **macOS:** `~/Library/Application Support/QGIS/QGIS3/profiles/default/python/plugins/`

Copy the entire `wetspass_modern` folder into that directory.

Open QGIS. Go to **Plugins → Manage and Install Plugins → Installed**. You should see **WetSpass-M Modern** listed. Ensure its checkbox is ticked.

A toolbar icon and a **WetSpass-M Modern** menu entry appear. If the icon is missing, verify that `icon.png` is present in the plugin folder.

2.3 Verifying dependencies

If `rasterio` is missing:

- **Windows:** open the OSGeo4W shell (installed with QGIS) and run: `pip install rasterio`
- **Linux / macOS:** use the same Python interpreter QGIS uses; run: `pip install --user rasterio`

If `geopandas` is missing (required for the crop tool):

Text: pip install geopandas

2.4 Shipping the user guide

The Help tab looks for this PDF in the plugin folder. Place it at:

wetspass_modern/help/WetSpassM_UserGuide.pdf

If it is not present, the Help tab shows a message with the expected path, but the quick-reference material remains available.

3. Quick start in five minutes

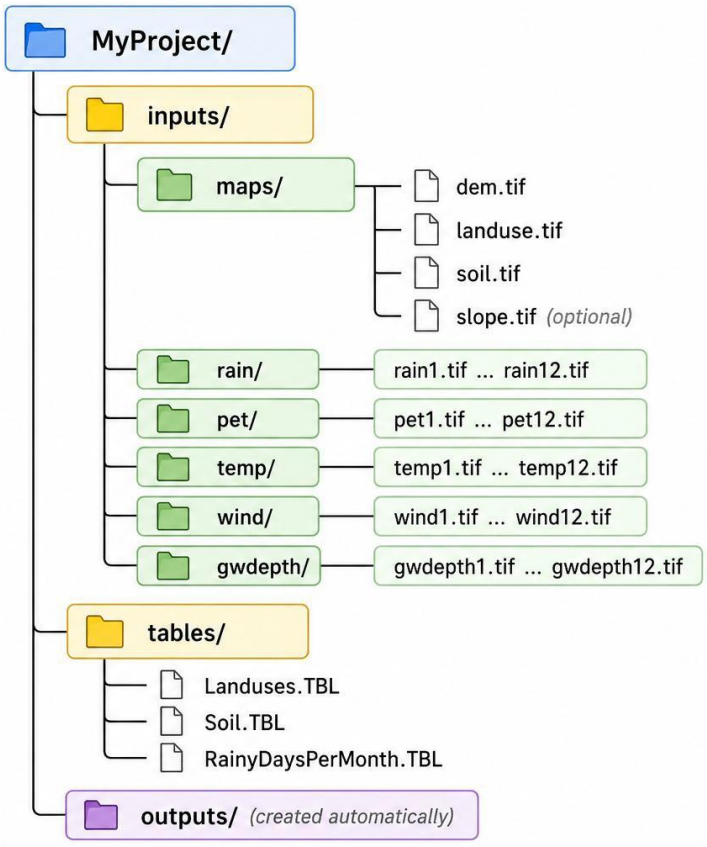
This is the fastest path from zero to a valid annual water balance. Details for each step follow in later chapters.

1. Prepare your input folder using the standard layout shown in section 4.
2. Open the plugin: **WetSpass-M Modern → Open**.
3. **Inputs tab.** Set the working directory. Click **Auto-load**. Confirm the DEM, land use, and soil paths are filled.
4. **Check Inputs tab.** Press **Check Inputs**. Every row should be green. Fix any red rows before continuing.
5. **Parameters tab.** Leave the defaults (LP = 0.85, Alfa = 1.5) unless you have local calibration data.
6. **Run tab.** Set the output directory. Leave **Spin-up cycles = 2** and **Compatibility mode = OFF**.
7. Press **Run**. Watch the Progress tab. The workflow reports convergence after each spin-up cycle and exits early when the bucket is stable. If you are satisfied that the bucket has converged and do not want to wait for the spinner count, press **Skip to final cycle**.
8. **Water Balance tab.** Click **Load from output folder**. The closure error should be close to zero.
9. *(Optional)* **Focused Recharge tab.** Click **Recompute from inputs**, run focused recharge, then click **Combine → Total recharge**.
10. *(Optional)* **Crop outputs to boundary.** Click the button on the tools row, tick the rasters you want, and confirm.

4. Input data requirements

4.1 Standard folder layout

All inputs live inside the working directory. The recommended layout is:



The **Auto-load** button in the plugin scans this layout and fills the paths for you.

Alternative layouts are also recognised (`maps/` + `tables/`, `Inputs/Maps/` + `Inputs/Tables/`, and `Maps/` + `Tables/`). The Check Inputs tab tells you which layout it detected.

4.2 Grid alignment

All raster inputs must share exactly the same grid as the DEM: CRS, pixel size, origin, width, and height. Resolution alone is not enough — two 30 m grids with different origins are still misaligned.

To align rasters in QGIS: **Raster → Projections → Warp**. Set the target CRS and extent to the DEM, use bilinear for continuous variables (rain, temperature, wind, PET, gwdepth) and nearest for categorical variables (land use, soil).

The Check Inputs tab runs this alignment test automatically.

4.3 Raster formats

GeoTIFF is preferred. Legacy ASCII `.asc` grids are accepted for backward compatibility. Output is always GeoTIFF.

4.4 Units

Variable	Unit	Typical range
Rainfall	mm/month	0 – 500
PET	mm/month	0 – 300
Temperature	°C	−30 – 45
Wind speed	m/s	0 – 15
Groundwater depth	m below surface	0 – 200
DEM	m	project-dependent
Slope	percent	0 – 100

All fluxes are monthly totals, not intensities. Do not supply daily rates.

4.5 Monthly file naming

The model expects a 12-file pattern per climate variable:

rain1.tif … rain12.tif
pet1.tif … pet12.tif
temp1.tif … temp12.tif
wind1.tif … wind12.tif
gwdepth1.tif … gwdepth12.tif

Prefixes are configurable on the Settings tab.

4.6 Lookup tables

Three tables are read from `inputs/tables/`.

Landuses.TBL — tab-delimited. One row per land-use class code.

Column	Meaning
1	Land-use code (integer)
4	Vegetated fraction (0–1)
5	Bare-soil fraction (0–1)

6	Impervious fraction (0–1)
7	Open-water fraction (0–1)
8	Rooting depth (m)
9	Leaf area index (LAI)
10	Minimum stomatal resistance (s/m)
11	Vegetation roughness height z0 (m)
13	Land-use factor for runoff coefficient
14	Aerodynamic resistance at 1 m/s wind (s/m)

Soil.TBL

Column	Meaning
1	Soil code
3	Field capacity (fraction)
4	Wilting point (fraction)
6	Residual water content (fraction)
7	Soil moisture parameter a
8	Evaporation depth (m)
9	Tension height at air-entry (m)
12	Soil factor for runoff coefficient
13	Saturated Hydraulic Conductivity (Ks in m/s)

RainyDaysPerMonth.TBL — Two columns: month number (1–12), mean number of rainy days per month.

4.7 Optional inputs

- Snow cover (`snowcover1.tif` … `snowcover12.tif`): fraction 0–1. Enable on the Run tab.
- Irrigation (`irrig1.tif` … `irrig12.tif`): additional water applied to the soil.
- LAI per month in `inputs/maps/LAI/`.

5. Tab-by-tab reference

5.1 Inputs tab

Sets the working directory and the primary rasters. The Auto-load button scans the standard folder layout and fills the paths automatically.

Fields:

- **Working Directory** — root of the project, contains `inputs/` and `outputs/`.
- **DEM map, Land use map, Soil map, Slope map** — paths to the static rasters. If Slope is left empty and "Create slope map from DEM directly" is ticked, the model computes slope internally.
- **Project boundary shapefile** — polygon used by the Crop outputs to boundary tool.
- **No of time steps** — usually 12 for a full year.
- **Starting / Final time step** — usually 1 and 12.
- **Initial discharge** — initial surface and subsurface flow values for the surface-water interaction option. Only used if the model is configured with basin area > 0.

5.2 Check Inputs tab

Runs a full validation:

- Every required lookup table exists and has enough columns.
- Every monthly raster exists for every requested month.
- Every raster shares the DEM's CRS, transform, width, and height.

Results are colour-coded. Red rows block the run — fix them first. Orange rows (grid mismatch) are warnings that usually indicate the model will produce garbage, so they should be fixed too.

Only explicit Check Inputs clicks write to the run log. Auto-checks triggered by switching tabs stay silent, so the log is not filled with duplicate lines.

5.3 Settings tab

Configures file-name prefixes for each variable. If your files are named `precip1.tif` instead of `rain1.tif`, change the prefix here. Also configures the nodata value (default −9999).

5.4 Parameters tab

Calibration coefficients.

Parameter	Default	Notes
"a" interception	4.5 mm/day	Interception threshold
Alfa	1.5	Priestley-Taylor coefficient
LP	0.85	Soil-moisture limitation
Average intensity	4 mm/h	Used in runoff formula
Slope factor	0.4	Weight in runoff coefficient
Land-use factor	0.3	Weight in runoff coefficient
Soil factor	0.3	Weight in runoff coefficient
Basin area	10 km²	For surface-water interaction
"x" coefficient	0.5	Surface-water interaction
Beta	0.75	Baseflow coefficient
Contribution factor	0.5	Recharge contribution

5.5 Output Period tab

Choose which temporal aggregations to write:

- **Monthly** — 12 files per variable.
- **Seasonal** — wet-season and dry-season files.
- **Annual** — one file per variable.

Season definitions are preloaded for several climate regions; the Ethiopian preset uses Kiremt (June–September) as wet season and the rest as dry season.

5.6 Run tab

- **Output Directory** — where rasters and reports go.
- **Model version** — compatibility mode checkbox (see section 12).
- **Spin-up cycles** — number of extra warm-up passes before the reported pass (see section 7). Default 2.
- **Save final soil moisture** — writes `soilwater_storage_initial.tif` for use as the starting state in future runs.
- **Reset soil moisture state** — deletes the state file, forcing the next run to start from a dry profile.
- **Enable automatic checkpointing** — writes a checkpoint after every completed month, so a power loss or drive disconnection resumes where it stopped.

- **Create simulation file** — writes `Simulated.tbl` with per-month mean fluxes.
- **Include snow processes** — enables the snow module.

5.7 Progress tab

Live log and progress bar. Controls:

- **Skip to final cycle** — abandons the remaining spin-up cycles and jumps straight to the final output cycle. Use this once the bucket has essentially converged and you do not want to wait for the spinner count. The current cycle finishes cleanly, then the workflow moves to the final pass.
- **Pause** — stops at the next block write and saves a checkpoint. The next run can resume from it.
- **Cancel** — stops at the next block write without saving a checkpoint. Partial output remains on disk until the next fresh run.
- **Close dialog** — hides the window; the run continues in the background.

5.8 Results tab

Choose which intermediate rasters to keep on disk. By default the model keeps monthly recharge and AET rasters; the rest are deleted after aggregation to save space.

5.9 Water Balance tab

See section 11.

5.10 Focused Recharge tab

See section 9.

5.11 Crop outputs to boundary

See section 10.

5.12 About tab

Scientific justification, equations, references. See section 6 for a summary.

5.13 Help tab

Opens this PDF and provides a quick-reference of the workflow, output folder structure, and common errors.

5.14 Calibration tab

Reads an observed CSV and the run's `Simulated.tbl`, computes NSE, KGE, PBIAS, R^2, RMSE, and MAE, and displays a time-series comparison. See Chapter 13 for a full explanation of the metrics and a recommended calibration workflow.

6. Physics and equations

6.1 Water balance at a cell, per month

The model partitions precipitation into four fluxes and one storage change:

P = AET + SR + R + ΔS

where:

- **P** = precipitation (rainfall plus snowmelt)
- **AET** = actual evapotranspiration to the atmosphere
- **SR** = surface runoff
- **R** = recharge to the saturated zone
- **ΔS** = change in soil-water storage during the month

6.2 Interception

$$a_aii = a \cdot LAI$$
$$Va = P - (1 - 1/\exp(0.463 - LAI))$$
$$DIT = a_aii - a_aii / (1 + Va / a_aii)$$
$$I_total = vegarea \cdot P \cdot (1 - \exp(-DIT \cdot nr / P))$$
$$I_total = \min(I_total, PET)$$
$$I = I_total - (0.04 + 0.067 \cdot LAI)$$

where nr is the number of rainy days in the month, a is the interception parameter (default 4.5 mm/day), and $vegarea$ is the vegetated fraction of the cell.

The factor $(0.04 + 0.067 \cdot LAI)$ splits total interception into a portion that evaporates directly (which is the flux I that enters AET) and a remainder that drips back to the soil.

6.3 Surface runoff

Runoff is computed with a modified runoff-coefficient method:

$$Cper = w_lu \cdot LU_factor + w_soil \cdot Soil_factor + w_slope \cdot slope / (10 + slope)$$
$$Cimp = 0.09 \cdot \exp(2.4 \cdot (imp + ow))$$
$$Cwp = (1 - imp - ow) \cdot Cper + (imp + ow) \cdot Cimp$$
$$Csr = (Cwp \cdot Pd) / (Cwp \cdot Pd - I_coef \cdot Cwp + I_coef) \text{ if } I_coef < Pd \text{ Cwp otherwise}$$
$$Q = Ch \cdot Csr \cdot (P - I_total)$$

where $Pd = P / nr$ is the average daily rainfall, $I_coef = 24 / I_avg$ is the inverse of the average rainfall intensity, and Ch is a soil-moisture-driven runoff modifier.

6.4 Actual evapotranspiration

AET is the total flux of water from the land surface to the atmosphere. It combines six physically distinct components:

AET = I + E_transp + E_bare + E_imperv + E_gw + E_ow

where:

- **I** = interception that evaporates directly from vegetation surfaces
- **E_transp** = transpiration from soil moisture in the root zone
- **E_bare** = evaporation from bare soil
- **E_imperv** = evaporation from impervious surfaces
- **E_gw** = transpiration and evaporation drawn from a shallow water table
- **E_ow** = evaporation from open water bodies

The soil-derived components are computed with a soil-moisture limitation:

$$E_transp = PET_veg \cdot (1 - a \cdot (awc / PET_veg)) \cdot evap_rate$$
$$E_bare = PET \cdot (1 - a \cdot (awc / PET)) \cdot evap_rate$$

where `awc` is available water capacity plus soil storage, and `a` is the soil moisture parameter from Soil.TBL.

In dynamic mode, the soil-water supply is `S_prev + P − I − SR − E_imperv`, not just the current month's rainfall. This is the key difference from 2016: the bucket carries moisture forward so that a wet month can supply transpiration in a subsequent dry month. The groundwater-fed component (E_gw) is computed separately and reported in the diagnostic raster `Cell_gw_discharge`. This raster is provided for interpretation — it tells you how much of the total AET was drawn from the water table rather than from the soil column. **It is already included inside AET**, so it must not be subtracted a second time in the closure equation.

6.5 Recharge

Recharge is the overflow from the soil bucket:

$$S_max = (fc - wp) \cdot rootdepth \cdot 1000 \quad [mm]$$
$$A = S_prev + P - I - SR - E_imperv$$
$$E_s = \min(E_s_potential, A) \quad [tp + bare]$$
$$S_new = clamp(A - E_s, 0, S_max)$$
$$R = \max(0, A - E_s - S_max)$$

`fc` and `wp` come from Soil.TBL, `rootdepth` from Landuses.TBL.

6.6 Snowmelt

Snow is tracked as snow water equivalent (SWE):

$$SWE = SWE_prev + P / (1.61 - 1.35 \cdot T + 1)$$
$$Cm = \text{melt factor if } T > \text{base temperature, else } 0$$
$$Melt = \min(Cm \cdot degree_days \cdot T, SWE)$$
$$SWE = SWE - Melt$$

`degree_days` is read from DegreeDaysPerMonth.TBL.

7. Spin-up and convergence

7.1 Why spin-up is needed

Because soil moisture is now a state variable, the model needs to run the same 12 monthly input maps several times before the bucket reaches a repeating annual cycle: S at the start of year N equals S at the start of year N + 1 at every cell.

Only then does the reported year have a well-defined mass balance and a ΔS that can be compared against P.

7.2 Recommended spin-up counts

Climate	Cycles
Maritime temperate	1
Continental / Mediterranean	2
Monsoon (Ethiopia, India, West Africa)	2–3
Arid / semi-arid	3–5
Default (may auto stop on spin up convergence)	30

The spinner on the Run tab sets extra warm-up passes before the reported pass. For instance, Set to 7 → 8 passes total.

In general, **Spin-Up Cycles (Default: 30):** The model repeatedly runs the 12-month cycle until the soil moisture stabilizes. Because the workflow is programmed to **exit early** the moment convergence is achieved, a generous default of 30 cycles costs absolutely nothing in terms of computation time. It simply ensures the model has enough runway to stabilize even in the most complex, arid, or high-altitude catchments that might take 10 to 15 cycles to reach equilibrium.

7.3 Automatic convergence check

After every spin-up cycle the workflow computes:

mean |S_end − S_start| [mm/yr]

If this falls below a tolerance, the bucket has converged and the remaining warm-up cycles are skipped automatically. The default tolerance is:

tolerance = max(1.0 mm, 0.5% × mean annual P)

For P = 1170 mm/yr, the tolerance is 5.86 mm/yr. A 5% margin is applied on top, so values within 5% of the tolerance also count as converged. This prevents the workflow from running an extra cycle when the residual is sitting on the numerical noise floor.

7.4 Convergence report

Every run that has at least one spin-up cycle writes `spinup_convergence.csv`:

```
Spin-up convergence report
mean_annual_P_mm,1171.6800
tolerance_mm,5.8584
tolerance_frac,0.005000
converged_after_cycle,3
cycle,mean_abs_dS_mm,mean_dS_mm,pct_of_P,tol_mm,converged
1,47.210314,41.305442,4.031023,5.858400,no
2,8.417295,7.441808,0.718202,5.858400,no
3,1.208741,0.918376,0.103163,5.858400,yes
```

Interpretation:

- `mean_abs_dS_mm` — the convergence metric for that pass.
- `pct_of_P` — the metric as a fraction of mean annual P.
- `converged` — yes if the metric fell below the tolerance (with margin).

If `converged_after_cycle` equals the spinner value, the bucket was still drifting at the last allowed pass. Add 1–2 cycles and re-run.

When spin-up is set to 0, no spin-up cycle runs and the CSV is not written. The final cycle runs directly from the state stored in `soilwater_storage_initial.tif`.

7.5 Manual verification

Two independent checks confirm convergence:

- In the Water Balance tab, ΔS is close to zero — within the tolerance reported in the Progress log.
- The last row of `spinup_convergence.csv` has `converged = yes`.

If both checks pass, the reported mass balance is trustworthy.

8. Outputs and interpretation

8.1 Output folder structure

outputs/

```
├── Lookups/                          static lookup rasters
├── Interception_1.tif … _12.tif      monthly outputs
├── Cell_runoff_1.tif … _12.tif
├── Cell_evapotranspiration_1.tif …
├── Recharge_diffusive_1.tif …
├── Recharge_diffusive_annual.tif     annual aggregate
├── Recharge_diffusive_wet_season.tif   seasonal (if enabled)
├── Recharge_diffusive_dry_season.tif   seasonal (if enabled)
├── Recharge_focused_annual.tif       (after Focused Recharge tab)
├── Recharge_focused_annual_capped.tif  (optional, if cap enabled)
├── Recharge_Total_annual.tif         (after Combine button)
├── cropped_to_boundary/              (after Crop button)
├── soilwater_storage_initial.tif     bucket state for next run
├── soilwater_storage_delta_annual.tif  ΔS over the reported year
├── spinup_convergence.csv            spin-up report (when spin-up > 0)
├── wetspass_checkpoint.json          resume state (deleted on completion)
├── Simulated.tbl                     per-month mean fluxes
├── RunSummary.txt                    human-readable run report
└── _run_metadata.json                machine-readable run metadata
```

8.2 Output variables

Name	Meaning	Units
Interception	Interception of the cell	mm/month
total_Interception	Interception before the 0.04 + 0.067·LAI factor	mm/month
evap_rate	Evaporative efficiency factor (0–1)	–
Cell_runoff	Runoff of the cell	mm/month
Cell_evapotranspiration	AET of the cell	mm/month
Cell_gw_discharge	Groundwater + open-water part of AET	mm/month
recharge	Diffuse recharge	mm/month
vegrunoff	Runoff in vegetated area	mm/month
barerunoff	Runoff of bare soil	mm/month
imperrunoff	Runoff of impervious area	mm/month
owrunoff	Runoff on open water	mm/month
Cell_actualtranspiration	Total actual transpiration	mm/month
Cell_actual_baresoil_evapo	Evaporation from bare soil	mm/month
Cell_ow_evaporation	Evaporation from open water	mm/month

Name	Meaning	Units
Cell_imper_evaporation	Evaporation from impervious surface	mm/month
Cell_gw_transpiration	Groundwater transpiration	mm/month
Cell_gw_evaporation	Groundwater evaporation	mm/month
penmann_coefficient	Penmann coefficient	–
wb_error	Per-cell water balance closure error	mm/month

8.3 Reading the annual rasters

Annual rasters are the sum of the 12 monthly rasters. For continuous variables (rain, ET, runoff, recharge) this is correct. For state variables (`soilwater_storage`), annual aggregation is not meaningful and the model writes them as monthly snapshots instead.

8.4 Simulated.tbl

Text file with the catchment-mean fluxes per month:

```
TimeStep  AET      Runoff   Interception  Recharge
1         23.4567  12.3456  5.6789        8.9012
2         25.1234  11.2345  6.7890        7.8901
...
```

Useful for calibration against observed annual hydrographs. When a run is resumed, the file is appended to rather than truncated, so months written before the pause are preserved.

9. The Focused Recharge module

9.1 What it computes

Focused recharge is the streambed infiltration along ephemeral or intermittent streams. It is a separate process from diffuse recharge and typically concentrated on a small fraction of the catchment area.

The module:

- Derives a stream network from the DEM (D8 flow accumulation).
- Routes monthly surface runoff downstream.
- Applies an exponential transmission loss on streambed cells.
- Reports the fraction of the network treated as losing vs gaining.
- Applies a climate-zone-based physical cap on the combined total recharge.

9.2 Parameters

- **Stream threshold** — accumulation threshold in cells (typical 500).
- **K_s** — streambed hydraulic conductivity, either from Soil.TBL or a fixed value. Typical clogged values: 1e-7 to 1e-6 m/s.
- **Channel width** — from Strahler order, from a shapefile attribute, or fixed.
- **Losing / gaining threshold** — depth-to-water-table cutoff. Streambed cells with a shallower water table are treated as gaining and skip loss.
- **Flow duration fraction** — fraction of each rainy day during which ephemeral streams actually flow. Typical 0.20–0.40.
- **Maximum infiltration depth** — physical cap, typical 20 mm/day.

9.3 Climate regime

The climate regime is auto-selected from rainfall when you click **Recompute from inputs**:

Mean annual P	Regime
< 300 mm/yr	Arid
300 – 700	Semi-arid
700 – 1200	Mixed / transitional
≥ 1200	Humid

Rainfall is more reliable than depth-to-water-table for this classification, since gwdepth rasters are typically interpolated from sparse borehole observations and can easily be off by tens of metres. The gwdepth-based classifier remains available as a secondary cross-check via the **Auto-detect from gwdepth (secondary)** button.

9.4 Combine methods

When you click **Combine → Total recharge**, you choose how the diffusive and focused rasters are combined into a single total recharge raster. The choice matters because the two estimates represent different physical pathways to the water table and therefore need to be combined without double-counting and without discarding a real contribution.

Addition (default). `R_total = R_diffusive + R_focused`. This is correct when the two mechanisms are physically independent pathways, which is the standard WetSpass-M formulation: a water molecule that infiltrates through the soil matrix is counted once by the diffusive model, and a water molecule that reaches a streambed and infiltrates there is counted once by the focused model. Addition is the recommended default for operational studies.

Replace-by-max (conservative). `R_total = max(R_diffusive, R_focused)`. At each cell the larger of the two estimates is kept. This avoids any chance of double-counting, at the cost of a systematic underestimate at every stream cell by `min(R_diff, R_foc)`. Use this only if a reviewer or a legacy comparison requires the conservative assumption, and document the choice.

Focused-over-diffusive. `R_total = R_focused` on stream cells, `R_diffusive` elsewhere. Use this only if you believe the diffusive model already includes some streambed loss and you want to replace that component with the focused estimate rather than add to it.

The chosen method is stored in the tab between runs. The log and the summary dialog both report which method was used, so the provenance of any given `Recharge_Total_annual.tif` file is unambiguous.

9.5 Physical cap

The focused-recharge engine can produce extreme values at individual stream cells when upstream runoff is concentrated into a small streambed area. Even with the mass-balance and monthly depth caps that the engine applies internally, the combined total can occasionally exceed what is physically plausible for a given climate. To guard against this, the combine step offers an optional physical cap derived from mean annual precipitation.

Clicking **Recompute from inputs** reads the monthly rainfall rasters, computes mean annual precipitation, classifies it against the standard reference table, and sets the recommended cap to the upper bound of the plausible recharge fraction for the matched zone.

Reference table for the physical cap:

Climate zone	Mean annual P (mm/yr)	Recharge fraction	Recommended cap (mm/yr)
Arid	100 – 300	1 – 5 %	5 – 15
Semi-arid	300 – 700	5 – 15 %	15 – 100
Humid temperate	700 – 1500	20 – 35 %	150 – 500

Climate zone	Mean annual P (mm/yr)	Recharge fraction	Recommended cap (mm/yr)
Wet tropical	1500 – 3000	30 – 50 %	450 – 1500
Extreme cloud forest / karst	3000 – 5000	40 – 60 %	1200 – 3000

The cap is optional and can be overridden. The recommended value is pre-filled into the override spinbox so you can see the number the plugin is suggesting and either accept it or adjust it.

The **Preview cap effect** button reports, without writing anything to disk, exactly what a given cap would do. It scans the two combine inputs tile-by-tile and reports the number of valid cells, before-cap and after-cap min/mean/max statistics, the number of cells that would be clipped, the mean drop across all cells, the overall max reduction, and a binned distribution of the clipped values by ratio to the cap.

When the cap is enabled and the **Also write capped focused raster** box is ticked, a separate file `Recharge_focused_annual_capped.tif` is written alongside the total. The original `Recharge_focused_annual.tif` is never overwritten.

9.6 Workflow

A typical focused-recharge run proceeds as follows:

1. Run the main (diffusive) model first. The focused engine reads `Cell_runoff_*.tif` from the output folder and cannot proceed without them.
2. Open the Focused Recharge tab.
3. Click **Recompute from inputs**. This computes mean annual precipitation from the monthly rain rasters, auto-selects the climate regime, and fills the recommended physical cap.
4. Configure parameters as needed. The **Auto-detect from gwdepth (secondary)** button is available if you want a cross-check.
5. Press **Run focused recharge**.
6. Optionally press **Preview cap effect** to check how the cap will behave before committing to it.
7. Press **Combine → Total recharge** to write `Recharge_Total_annual.tif`.
8. Return to the Water Balance tab and click **Update recharge**.

The workflow is checkpointed after every month, so if the run is interrupted, clicking Run again resumes from the last completed month rather than starting over.

10. Crop outputs to boundary

10.1 What it does

The **Crop outputs to boundary** button on the tools row removes the rectangular "zero collar" that the DEM alone leaves behind by cropping every selected output raster to a polygon boundary. It replaces the failing QGIS `gdal:rasterize` workflow.

10.2 Why it exists

The QGIS "Mask rasters" tool uses `gdal:rasterize` on the command line. It frequently fails with either `ERROR 1: Duplicate argument -ot` or a silent no-output when the boundary polygon's CRS differs from the raster's, or when the two layers do not spatially overlap. WetSpass-M Modern's crop tool bypasses the GDAL command-line layer and uses `rasterio` plus `geopandas`, which handle CRS reprojection and boundary-pixel retention correctly.

10.3 Selection dialog

Clicking the button opens a dialog listing every `.tif` file in the output folder, grouped by variable and period. Tick the ones you want cropped.

Controls:

- **Filter** — type a substring to narrow the list.
- **Select all,** Select none**,** Invert**.**
- **Only annual** — ticks only `*_annual.tif`.
- **Only Recharge** — ticks only `Recharge_*.tif`.
- A running counter shows how many of the available rasters are selected.

10.4 Where the output goes

Cropped files are written to a subfolder named `cropped_to_boundary/` inside the source output folder. The originals are never modified, so you can compare cropped and uncropped versions of any file side by side.

10.5 Behaviour

The tool auto-reprojects the boundary vector to each raster's CRS, caching the reprojection per unique CRS. Edge pixels are retained with `all_touched=True`, which is critical for hydrological water-balance closure — discarding edge cells systematically biases catchment-scale recharge. Pixels outside the boundary are set to nodata.

The crop runs on a background thread. The GUI stays responsive while a large folder is being processed, progress is reported in the tab's progress bar, and every file — success or failure — is

logged with its name and the resulting pixel dimensions. A failed file does not stop the rest of the queue.

10.6 Recommended workflow

1. Produce all the rasters you want to keep: diffusive, focused, total, and any annual or seasonal aggregates.
2. Set the project boundary shapefile on the Inputs tab.
3. Click **Crop outputs to boundary**.
4. In the dialog, tick the rasters you want cropped. Typically the annual files plus any seasonal files.
5. Click OK. The originals remain in place; the cropped copies are written to `cropped_to_boundary/`.
6. Load the cropped rasters from `cropped_to_boundary/` into QGIS for visualisation or further analysis.

11. The water balance panel

11.1 Reading the panel

The panel shows the annual average of each flux and the closure error:

Mass balance: P = AET + SR + R_total + ΔS
Closure error = P − AET − SR − R_total − ΔS
The colour of the closure error is green if |error| is within the acceptable band (< 1% of P), amber if 1–5%, and red above 5%. A green closure on a spun-up run is the strongest evidence that the model has produced a physically consistent water balance.

11.2 What ΔS means

ΔS is the change in stored soil moisture over the reported year. On a converged run it is close to zero — typically under the tolerance of 0.5% of P. If ΔS is larger, the bucket has not yet reached a repeating annual cycle. Increase spin-up cycles by 1 and re-run.

ΔS is not a closure-error correction. It is a real, physical flux that belongs in the balance and would be non-zero even in a perfect model run, because the soil column still accumulates or releases some water during the reported year.

11.3 Focused recharge integration

If you have run the Focused Recharge module, the panel reads `Recharge_focused_annual.tif` and adds it to `R_total`. If you have also produced `Recharge_Total_annual.tif` with the Combine button, the panel reads that file directly. The panel distinguishes these cases in the status line beneath the buttons.

12. Compatibility mode

12.1 What it does

Ticking **Wetspass-M 2016 compatibility mode** on the Run tab disables the dynamic bucket and restores the original physics:

- Soil moisture is reset to zero every month ($\Delta S = 0$ internally).
- Soil-derived ET is capped at rainfall, not at (P − runoff).
- Groundwater-fed ET is folded back into `Cell_evapotranspiration`.
- The recharge-driven ET adjustment is reapplied so that the closure error matches the original 2016 formulation.

12.2 When to use it

Use it to validate the modernized code against a legacy Wetspass-M 2016 installation on identical inputs. Cell-by-cell differences should be at the level of floating-point round-off. If they are larger, the implementation has diverged and should be investigated.

Do not use compatibility mode for operational studies in seasonally concentrated rainfall regimes — you will lose the seasonal redistribution that the bucket provides.

13. Calibration and validation

13.1 What the two words mean in this context

Calibration is the process of adjusting the model's tunable parameters so that its outputs reproduce observed data at a specific site. The parameters that can be calibrated are the ones on the Parameters tab: `a` (interception), `Alfa` (Priestley-Taylor coefficient), `LP` (soil-moisture limitation), `intensity`, `w_slope`, `w_landuse`, `w_soil`, and — if surface-water interaction is enabled — `x`, `Beta`, and `contribution`.

Validation is the process of testing a calibrated model against an *independent* dataset that was not used in the calibration. A model that reproduces its calibration data but fails on independent data is overfitted and should not be used for prediction.

Both activities are supported by the **Calibration tab** in the plugin.

13.2 What the Calibration tab does

The Calibration tab reads two inputs:

1. **Observed data** — a CSV or text file with at least two columns: time step (month number) and observed value. The plugin recognises the standard column names `timestep`, `time`, `month`, `date` for the first column, and `observed`, `obs`, `value`, `streamflow`, `runoff` for the second. If your column names differ, use the dropdown in the tab to map them.
2. **Simulated.tbl** — the per-month summary file written by the run itself, located in the output folder.

The tab then reports four goodness-of-fit metrics: NSE, KGE, PBIAS, and R^2, plus two absolute-error metrics: RMSE and MAE. It also draws a time-series comparison plot and a scatter plot of observed versus simulated values.

13.3 The metrics

NSE (Nash-Sutcliffe Efficiency) is the standard hydrology metric. It ranges from $-\infty$ to 1. Values above 0.75 are considered very good, 0.5 to 0.75 is acceptable, and below 0.5 indicates the model is not capturing the observed variability.

$$NSE = 1 - \Sigma(Q_{obs} - Q_{sim})^2 / \Sigma(Q_{obs} - Q_{obs_mean})^2$$

KGE (Kling-Gupta Efficiency) decomposes the error into three components: correlation, bias, and variability. A KGE of 1 is perfect; values above 0.75 are considered good; values below 0.5 indicate a problem. KGE is often preferable to NSE because it penalises variance errors, not just mean errors.

$$KGE = 1 - \sqrt{[(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2]}$$

where r is Pearson correlation, α is the ratio of standard deviations, and β is the ratio of means.

PBIAS (Percent Bias) measures systematic over- or under-prediction. A PBIAS of 0 is perfect. Positive values indicate the model over-predicts; negative values indicate it under-predicts. Values within ±10 % are usually considered acceptable.

$$PBIAS = 100 \cdot \Sigma(Q_{sim} - Q_{obs}) / \Sigma(Q_{obs})$$

R^2 is the squared Pearson correlation coefficient. It measures linear association but does not penalise bias or scale errors, so it should be reported alongside NSE or KGE, not instead of them.

RMSE (Root Mean Square Error) and **MAE (Mean Absolute Error)** are absolute-error metrics in the same units as the observed data. They are useful for reporting but not for cross-site comparison, because their magnitude depends on the scale of the data.

13.4 What to calibrate against

The most common calibration targets in WetSpass-M workflows are:

Target	Source	When to use
Annual streamflow	Gauge records	The standard. The catchment-integrated runoff must match observed discharge.
Monthly streamflow	Gauge records	If monthly data exist. Tests seasonal timing, not just annual totals.
Actual ET	Eddy-covariance flux tower	If you have a flux tower in or near the catchment.
Groundwater levels	Observation wells	Indirect, but useful in heavily monitored basins.
GRACE-derived TWS	Satellite gravimetry	For large basins (> 100,000 km²). TWS change is a proxy for ΔS + groundwater change.

In most WetSpass-M applications, the calibration target is annual streamflow, and the tunable parameter with the strongest influence is LP (soil-moisture limitation), followed by Alfa.

13.5 Calibration workflow

A defensible calibration workflow proceeds as follows:

- ✓ **Split your observed record.** Reserve 30–40 % of the record for validation. For a 20-year record, use 12–14 years for calibration and 6–8 years for validation. Do not use the same data for both.
- ✓ **Start from the defaults.** Run the model once with the default parameters and record NSE, KGE,

and PBIAS against the calibration period.

- ✓ **Adjust `LP` first.** LP controls how much of the available soil moisture is lost to ET before recharge begins. Raising LP increases ET and reduces recharge; lowering it does the opposite. Adjust in increments of 0.05.
- ✓ **Adjust `Alfa` second.** Alfa scales the Priestley-Taylor PET. If your observed ET is systematically too low, raise Alfa; if it is too high, lower it.
- ✓ **Adjust the runoff weights last.** Only if runoff remains badly mismatched after LP and Alfa have been tuned. The three weights (`w_slope`, `w_landuse`, `w_soil`) must sum to approximately 1.0.
- ✓ **Re-run and re-evaluate.** Repeat until the metrics stabilise. Do not tune beyond three or four iterations — the model has limited degrees of freedom, and further adjustment is overfitting.
- ✓ **Validate on the held-out period.** Run the model with the calibrated parameters and compute NSE, KGE, and PBIAS on the validation period. If the validation metrics are much worse than the calibration metrics, you have overfitted and should back off the parameter adjustments.

13.6 Target values

The following ranges are widely used in the hydrological literature as thresholds for a defensible calibration:

Metric	Very good	Acceptable	Poor
NSE	> 0.75	0.50 – 0.75	< 0.50
KGE	> 0.75	0.50 – 0.75	< 0.50
PBIAS	± 10 %	± 10 – 25 %	> ± 25 %

If your calibration does not reach at least the "acceptable" band on the calibration period, the model is not representing the dominant processes at your site. Before tuning further, check:

- Are your inputs correct? A misaligned rainfall raster or a mis-specified land-use table will prevent any calibration from succeeding.
- Is the water balance closing? If the closure error is large, the calibration target is meaningless.
- Is the dominant process actually within the model? WetSpass-M is a monthly water-balance model. It does not simulate event-scale flooding, snow-dominated regimes with rain-on-snow events, or strongly karstic systems with rapid conduit flow.

13.7 Validation protocol

A complete validation reports the following:

1. **Held-out period.** Metrics computed on data not used in calibration.
2. **Multi-site validation.** If the model is to be transferred to ungauged basins, validate on several gauges.
3. **Sensitivity analysis.** Vary each calibrated parameter by ± 20 % and report how much the

metrics change. If the metrics are highly sensitive to a parameter that has no physical justification for being tuned, the calibration is fragile.

4. **Uncertainty range.** Report the range of parameters that give "acceptable" metrics, not just the single best-fit parameter set. A calibration that is narrow to a single point is usually overfitted.

13.8 Common calibration pitfalls

- **Tuning on the validation period.** This defeats the purpose of validation. Keep the two periods strictly separate.
- **Calibrating against the wrong variable.** If you calibrate runoff against a groundwater-level record, the calibration will be meaningless. Match the calibration target to the appropriate model output.
- **Ignoring the water balance.** A model that reproduces streamflow but has a closure error of 20 % is not a valid calibration, because it means the internal fluxes are wrong even though the outlet is right.
- **Over-tuning.** More than five or six parameter adjustments almost always means overfitting. Report the parameter set that gives the best validation, not the best calibration.
- **Skipping spin-up.** Calibration on a non-converged run is meaningless. Always let the bucket converge before comparing to observations.

14. Troubleshooting

14.1 The Check Inputs tab shows red rows

The tables or rasters listed are missing or malformed. Read the message column — it usually identifies the file path or the missing column.

- Missing file → check the file name spelling.
- Missing column → open the `.TBL` file and compare to the template.
- Grid mismatch → reproject the raster to the DEM grid.

14.2 "Input rasters are not on the same grid as the DEM"

This means CRS, transform, or size differs. Use **QGIS → Raster → Projections → Warp** to reproject every input to the DEM's grid. Verify the fix with the Check Inputs tab.

14.3 Recharge raster is all zeros

Three possible causes:

- The run did not complete — check the Progress log for errors.
- ET consumes all available water — check the Water Balance panel: if AET is close to P, recharge will be near zero. Adjust the LP or LAI parameters.
- The recharge raster was produced by an earlier buggy run — delete `Recharge_diffusive_*.tif` and re-run.

14.4 Closure error is large in the Water Balance panel

In order of likelihood:

- ΔS is large — the bucket has not converged. Increase spin-up cycles.
- Rasters are misaligned — Check Inputs will catch this.
- Compatibility mode was toggled between runs — discard the checkpoint and re-run.

14.5 Checkpoint errors on Resume

If a parameter, spin-up count, or compatibility setting changed after pausing, the checkpoint's parameter hash no longer matches. Click **Discard checkpoint** and start a fresh run.

14.6 Model appears stuck on a spin-up cycle

It is not stuck. Each cycle runs 12 months. Watch the "Month N — block M" lines. The workflow exits early when the bucket converges — look for the CONVERGED line in the Progress log. If you are satisfied with convergence and do not want to wait for the spinner count, press **Skip to final cycle**.

14.7 Power loss or drive disconnection during a run

Automatic checkpointing is on by default. The workflow saves its state after every completed month. When you reopen QGIS:

- The Run tab shows a "Paused run found" banner.
- The main button reads "Resume run" instead of "Run".
- Click Resume run and the workflow picks up at the exact month and block where it stopped.

Only one month of computation is ever at risk.

14.8 "Output device unavailable" warning

The external drive holding the output folder was disconnected. The workflow retried the write a few times and then paused cleanly. Reconnect the drive, then click Resume run on the Run tab. Only the month that was in progress will be recomputed.

14.9 Duplicate "[Check Inputs]" log lines

Fixed as of version 2.4.1. Auto-checks triggered by switching to the Check Inputs tab or changing the working directory no longer write to the run log; only an explicit Check Inputs click does.

14.10 Crop tool: "geopandas is not installed"

Install `geopandas` in QGIS's Python environment. On Windows, open the OSGeo4W Shell and run `pip install geopandas`. On Linux and macOS, use the same Python interpreter QGIS uses.

14.11 Crop tool: "Invalid path — Output directory does not exist"

Set the Output Directory on the Run tab before clicking the crop button.

14.12 Crop tool: some files fail to crop

The log line shows the file name and error. Common causes are the raster being open in QGIS and locked for writing, a boundary shapefile that is not readable, or a corrupted raster. Unload the raster, verify the boundary file, and re-run.

14.13 Focused recharge: "operands could not be broadcast together"

Fixed as of 2.4.1. The soil-derived Ks raster is now resampled to the D8 hydrology grid before use. If you see this on an older version, upgrade.

14.14 Focused recharge: outliers in Recharge_focused_annual.tif

The 2.4.1 engine caps each cell's monthly infiltration at `max_daily_depth_mm × flow_days` and enforces mass balance at every cell. If you still see large values, either the cap is too loose for your soils, the stream threshold is too low, or the DEM contains an artificial sink or spike. Adjust `max_daily_depth_mm`, `flow_duration_fraction`, or the stream threshold accordingly.

14.15 Combine: "Duplicate argument -ot"

Fixed as of 2.4.1. The width rasterization step no longer passes `-ot` twice.

14.16 Which combine method should I use

For standard WetSpass-M workflows, use **Addition**. Use **Replace-by-max** only if you have a specific reason to be conservative and are aware that it underestimates total recharge. Use **Focused-over-diffusive** only if you believe the diffusive model already includes some streambed loss.

14.17 The capped total looks too low

Check the recommended cap in the Focused Recharge tab. If it is much lower than you expect, click **Recompute from inputs** and verify the mean annual P is correct. Then use **Preview cap effect** to see how many cells are being clipped. If the cap is too aggressive for your catchment, tick **Override with** and enter a higher value.

15. Advanced Optional Features

WetSpass-M Modern includes three optional features designed to make the model faster, more memory-efficient, and more physically realistic in extreme climates. Depending on the size of your project and the climatic zone, these can be toggled on or off.

15.1 Auto-Cropping to Boundary

What it does: Automatically clips all input rasters (DEM, Landuse, Soil, and monthly Rain/PET/Temp) to the bounding box of your project boundary shapefile before the math begins. The cropped rasters are saved in outputs/cropped_inputs/ and reused if you resume the run.

When to Use (Recommended for most projects):

- **Large-scale projects** (e.g., regional, national, or >10,000 km²). When your input rasters are massive rectangles containing vast amounts of NoData (e.g., ocean, neighboring countries), the model wastes RAM and CPU trying to process dead pixels.
- **Limited RAM:** If your computer has less than 16GB of RAM, processing 100+ million pixels will trigger a MemoryError. Auto-cropping reduces the pixel count drastically.
- **External Hard Drives:** If working on a slow USB HDD, cropping to the boundary reduces the data volume, making subsequent read/write cycles significantly faster.

When to Opt Out:

- **Small-scale projects** (e.g., a single small watershed < 500 km²) where the raster's bounding box is already tightly clipped to the watershed, making the cropping step redundant.

Assumptions Considered:

- The shapefile provided accurately represents the hydrological boundary of the study area.
- The bounding box of the shapefile will contain all necessary data for edge pixels (the model does not require a buffer zone outside the boundary for this specific spatial calculation).
- The user has enough disk space to store the duplicated, cropped rasters.

15.2 Groundwater Depth (GWD) Capping (UNEP 5-Zone Climate)

What it does: Automatically caps the groundwater depth using the UNEP Aridity Index (calculated from your PET and Rainfall). It forces deep water tables in arid regions and allows shallow tables in humid regions, preventing "infinite evaporation" and unrealistic recharge estimations.

When to Use (Recommended for Arid/Semi-Arid regions):

- **Arid, Semi-Arid, and Dry Sub-Humid climates.** In these regions, the real water table is often 30–200 meters deep or even more. The model's soil bucket is usually only 1–3 meters deep. Without capping, the model assumes groundwater is touching the soil, causing massive over-evaporation and

artificially high recharge estimates.

- **When using global or regional datasets:** If your groundwater depth input is estimated (e.g., from a global model) rather than locally measured, capping protects the physics.

When to Opt Out:

- **Humid climates or Wetlands:** If the water table is genuinely shallow (< 2 meters) and interacts with the surface (e.g., peatlands or river valleys), capping could artificially disconnect it.
- **High-fidelity local data:** If you possess highly accurate, localized groundwater depth data and you want the model to use those exact values without any mathematical modification.

Assumptions Considered:

- The PET and Rainfall rasters accurately represent the macro-climate of the region.
- The capillary rise properties derived from your Soil.TBL are standard. (The manual override is available if your soil data is missing or non-standard).
- Groundwater depth beyond the calculated cap (e.g., >30m in arid zones) has zero physical impact on the surface water balance.

15.3 Spin-Up Convergence Criteria

What it does: The model repeatedly runs the 12-month cycle to stabilize the soil moisture bucket. It automatically stops early if the soil moisture change (ΔS) drops below a strict tolerance (0.5% of mean annual rainfall).

When to Use (Recommended for final reporting):

- **Final Model Runs:** Always let the spin-up run to convergence for final thesis or publication results. This ensures that the soil is in perfect equilibrium with the climate, yielding the most accurate absolute recharge and AET values.

When to Opt Out (Skip to Final Cycle):

- **Hyper-Arid Regions:** In regions that are Arid and Semi-Arid, rainfall is so erratic that soil moisture may never perfectly converge (the cycle-to-cycle change plateaus). Waiting 3 days for 7 cycles might only change the recharge by 0.1 mm/yr. If the spatial pattern is stable after 3 cycles, clicking "Skip to Final Cycle" saves time.
- **Rapid Testing/Calibration:** When you are tweaking parameters and just want to see the spatial output, skipping the spin-up saves hours.

Assumptions Considered:

- **Starting from Zero:** The model assumes the initial soil bucket is dry (0 mm). The spin-up cycle effectively "forgets" this artificial start.
- **Stationary Climate:** The model assumes the 12 months of input data represent a repeating climatological average (no long-term climate trends).

Note: Skipping the spin-up does **not** break the mass balance. The final cycle's ΔS is still calculated and closes the equation perfectly.

16. River-Aquifer Conductance & Focused Recharge (Experimental)

16.1 Introduction & Novel Contributions

Estimating river-aquifer exchange is critical for groundwater modeling (MODFLOW) and for identifying Groundwater Potential (GWP) zones in semi-arid environments. Traditional models require detailed measurements of riverbed thickness, hydraulic conductivity, and width for every river cell—a logistical impossibility for large catchments.

This plugin introduces an **experimental Field-Anchored Geomorphic Scaling methodology**. The accuracy and reliability of the approach remain to be established through future studies conducted at different spatial scales and across diverse catchments. Its key innovation is that it requires detailed field measurements only at the catchment outlet, while remote sensing data and DEM-derived geomorphic characteristics are used to extrapolate and spatially distribute the measured parameters upstream.

Novel Assumptions & Original Formulations:

The Outlet Expansion Index (OEI): We introduce the OEI to handle the complex Lake Chamo delta. Rather than applying the massively expanded delta width (*Wout*) to the entire river network, the OEI isolates this expansion to the outlet, preventing artificial over-estimation of upstream conductance.

Dynamic Streambed Incision: The model assumes the DEM represents the general valley floor. The streambed is calculated as a physical incision below this floor (*Zbed=DEM−Depth*).

Stream Typology Weighting: Recognizing that ephemeral streams (dry most of the year) behave differently than Intermittent and perennial rivers, the model allows for a typology multiplier (Perennial = 1.0, Intermittent = 0.8, Ephemeral = 0.5) to be applied to the conductance, reflecting the duration of water availability.

16.2 Governing Equations & Units

Reference Conductance (C_0), based on Darcy's Law for the reference outlet reach:

$$C_0 = \frac{K_0 \times W_0 \times L_0}{M_0}$$

- K_0: Riverbed hydraulic conductivity (m/day)
- W_0: Reference channel width (m)
- L_0: Reference reach length (m) *(Note: This should ideally match your MODFLOW cell size)*
- M_0: Effective riverbed thickness (m)
- **Units:** m^2/day

This is the foundational equation for the entire **River Conductance methodology**. It is a direct adaptation of **Darcy's Law** applied to a specific riverbed segment. Here is a detailed elaboration of every component, why it is structured this way, and how to obtain the values:

The Physical Meaning

Conductance (C*C*) measures how easily water can flow between the river and the aquifer. Think of it like a pipe: a wider, shorter pipe with highly permeable material will let a lot of water through (high conductance), while a narrow, thick pipe with clay will block water (low conductance). The formula calculates the conductance exactly at the point where you took your field measurements (the outlet).

***K*0 (Riverbed Hydraulic Conductivity):**

- **What it is:** The permeability of the riverbed sediment (e.g., sand vs. clay).
- **Unit:** meters per day (*m*/*day*).
- **Where to get it:** Field permeameter tests, slug tests, or laboratory grain-size analysis of the riverbed soil, or may be from text book references. High values (e.g., 10*m*/*day*) mean sand/gravel; low values (e.g., 0.001*m*/*day*) mean clay/silt.

***W*0 (Reference Channel Width):**

- **What it is:** The wetted width of the river channel at your reference point.
- **Unit:** meters (*m*).
- **Where to get it:** Field survey with a tape measure, or measured via Google Earth imagery. (Use the median of several measurements to be safe).

***L*0 (Reference Reach Length):**

- **What it is:** The length of the river segment you are modeling.
- **Unit:** meters (*m*).
- **Where to get it & Crucial Note:** This should exactly match your MODFLOW grid cell size! If your MODFLOW model uses 100m x 100m cells, set *L*0=100. If it uses 500m cells, set *L*0=500. This ensures the conductance units translate correctly into the model.

***M*0 (Effective Riverbed Thickness):**

- **What it is:** The thickness of the sediment layer (the clogging layer) between the river bottom and the aquifer.

- **Unit:** meters (*m*).
- **Where to get it:** Soil probing with a steel rod, shallow hand-augering, or geophysics. It acts as a resistor in the formula. The thicker the clay layer, the *lower* the conductance will be.

***M*0 in the denominator?**

Because it represents resistance. If you have a very thick layer of muddy sediment (*M*0=5*m*), water struggles to pass through. If you have a thin layer of clean gravel (*M*0=0.1*m*), water flows easily. By dividing by *M*0, a thicker riverbed mathematically drops the conductance value.

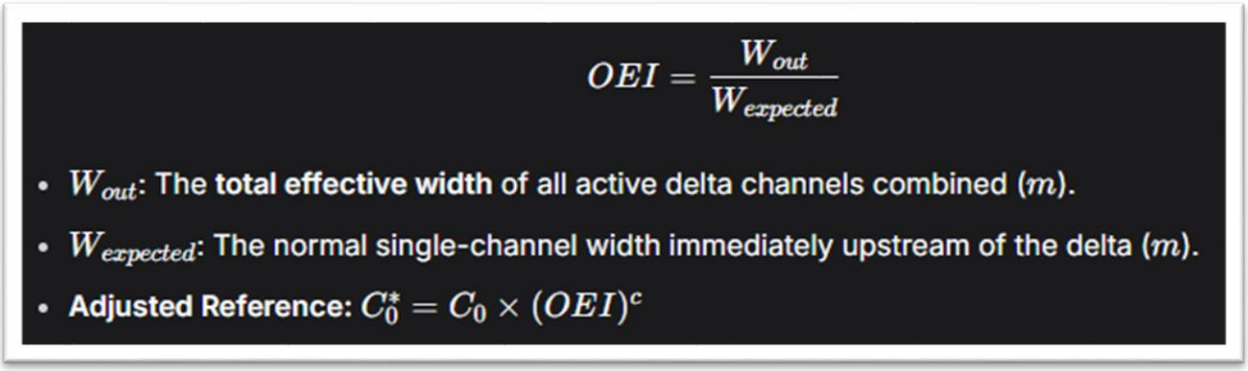
The Final Unit is in m2/day. This is the exact unit MODFLOW 6 expects for the River (RIV) package. It represents the volume of water (cubic meters) that exchanges per unit of head difference (meters) per day.

Example Calculation: If you measured: *K*0=1.5*m*/*day* (Sandy riverbed)**,** *W*0=20*m* (River width)**,** *L*0 =100*m* (Matching a 100m MODFLOW cell)**,** *M*0=0.5*m* (Half a meter of sediment)**, then** *C*0 =(1.5×20×100)/0.5)=**6,000 m2/day.**

The Next Step in Your Model:

Once you calculate *C*0, the model doesn't just apply 6,000 everywhere. It uses this value as the **anchor** at the outlet. Then, it uses the OEI (delta expansion) and the Strahler scaling (*Ai*,*Si*,*Ii*) to progressively lower this value as you move upstream into smaller and smaller tributaries.

Handling the multi-channel delta entering Lakes (Open Water Bodies):



The Outlet Expansion Index (OEI) is the novel core of the River Conductance methodology. It is specifically designed to solve a massive problem when modeling rivers that enter lakes or oceans. Here is a detailed breakdown of what the formula means, component by component:

The Problem It Solves: Standard models just take the width of the riverbed where you measured it (at the outlet) and apply it to the entire river network. But at a delta entering a lake, the river expands massively into multiple braided channels. If you apply that massive width upstream, the model thinks the entire river is 100 meters wide, which creates artificially huge conductance and "fake" water loss.

What is *Wout*? (Total Delta Width). The total effective width of *all* active channels where the river enters the lake. How to get it: If the river splits into three channels at the lake, each 15 meters wide, then *Wout*=15+15+15=45 meters. You measure this using Google Earth or field surveys.

What is *Wexpected*? (Normal River Width). The normal width of the single channel *just before* it starts expanding into the delta. How to get it: You look at your satellite imagery (field work) and measure the river a few hundred meters upstream of the widening (usually, where the normal River/Stream width starts). If the river is 15 meters wide there, W*expected*=15.

Spatial Conductance Propagation (C_i), applying Leopold & Maddock's (1953) hydraulic geometry theory:

$$C_i = C_0^* \times \left(\frac{L_i}{L_0}\right) \times \left(\frac{A_i}{A_0}\right)^b \times \left(\frac{S_i}{S_0}\right)^a \times \left(\frac{I_0}{I_i}\right)^d \times G_i \times T_i$$

- A_i, S_i, I_i: Drainage Area (km^2), Strahler Order, and Slope (m/m) for cell i.
- a, b, d: Sensitivity exponents (Fast-track defaults: $a = 0.5, b = 0.4, d = 0.4$).
- G_i: Geological factor (default 1.0).
- T_i: Stream typology multiplier.
- Units: m^2/day

If the previous formula (*C*0) gave you the conductance at the *single outlet point*, this formula (Spatial Conductance Propagation Formula, *Ci*) is the engine that scales that value up the entire river network. It calculates the conductance for every single river pixel (*i*) in your DEM. It relies on Leopold & Maddock's (1953) hydraulic geometry theory, which states that river width, depth, and flow scale predictably with drainage area and stream order.

The Base Component: *C*0∗ (Adjusted Reference Conductance):
This is the starting anchor calculated in the previous steps (Reference Conductance × Outlet Expansion Index). It represents the conductance at the delta/outlet.

The Scaling Ratios (The Math) - To calculate conductance at an upstream pixel, the model compares the upstream pixel's properties to the outlet's properties using ratios. All ratios are calculated as (Local Value / Outlet Value):

Li/*L*0 — Length Ratio: Scales linearly with the length of the river cell. *(Note: If your MODFLOW cells are all the exact same size, this ratio is always 1.0 and has no effect).*
$(Ai/A0)^b$ — Drainage Area Ratio: Drainage area is a proxy for how much water the river carries. A pixel draining 100 km² will have much higher conductance than a pixel draining 1 km².
$(Si/S0)^a$ — Strahler Order Ratio: Accounts for the discrete "step-jumps" in river size at confluences. A 4th-order stream has higher conductance than a 1st-order stream.

$(I0/Ii)^d$ — Inverse Slope Ratio: Notice the inversion. We divide the outlet slope ($I0$) by the local slope (Ii). Physically, this means that flatter river reaches have higher conductance. Why? Because water lingers longer in flat areas, allowing more time for it to infiltrate into the riverbed. Steep mountain streams flow too fast to lose much water.

The Modifiers (The Physics):

Gi (Geological Factor): Adjusts conductance based on the aquifer material. If a tributary flows over solid granite (low permeability), *Gi* would be e.g., 0.1. If it flows over clean alluvium, *Gi* would be e.g., 2.0. Default is 1.0.

How to decide on *Gi* (The Hydrogeological Logic):

The Geological Factor (*Gi*) represents how easily water can move through the aquifer material beneath the riverbed relative to a reference material. You determine it by looking at a geological map and assigning a relative permeability weight to each rock type:

Geological Material	Permeability	Suggested Gi*Gi* Value	Rationale
Clean Alluvium / Fluvial Sands	Very High	**2.0 - 3.0**	Water flows easily between the river and the aquifer.
Fractured Volcanic (Basalt)	High	**1.5 - 2.0**	Cracks and fractures hold and transmit water.
Colluvium / Weathered Rock	Moderate	**1.0**	Average transmission.
Massive Volcanic (Unfractured)	Low	**0.3 - 0.5**	Very hard rock; river water struggles to enter.
Metamorphic / Granite / Shale	Very Low	**0.1 - 0.2**	Almost impermeable; acts as an aquitard.

How to create the *Gi* Raster in QGIS

- Load your Geological Map (shapefile) into QGIS.
- Open the attribute table and add a new field called GEO_FAC (Type: Decimal).
- Fill in the values based on the table above (e.g., if a polygon is "Alluvium", type 2.0).
- Go to Raster > Conversion > Rasterize (Vector to Raster).
- Set the Input to your geology shapefile, the Field to GEO_FAC, and crucial step: set the Output Raster Size to exactly match your DEM grid (e.g., using the dem.tif as the reference).
- Save it as Geology_Factor.tif.

***Ti* (Stream Typology Multiplier): Adjusts based on flow duration:**

- Perennial: 1.0
- Intermittent: 0.8
- Ephemeral: 0.5

The Exponents (*a*,*b*,*d*)

These are the sensitivity knobs. They control how aggressively the conductance drops as you go

upstream.

- *a* (Order): Default 0.5
- *b* (Area): Default 0.4
- *d* (Slope): Default 0.4

Note: These defaults come from global averages of hydraulic geometry. You can adjust them in the UI to calibrate the model to your project area.

Units : The units output is *m*2/*day*.

Streambed Depth (D_i) & Stage ($Stage_i$):

$$D_i = D_0 \times \left(\frac{S_i}{S_0}\right)^f \quad \text{and} \quad Stage_i = Stage_0 \times \left(\frac{S_i}{S_0}\right)^g$$

- D_0: Measured depth at the outlet (m).
- $Stage_0$: Measured water stage above the bed at the outlet (m).
- f, g: Depth and Stage exponents (Fast-track defaults: 0.4).

In this formula, S_i **and** S_0 **refer to the Strahler Stream Order**. **What is Strahler Stream Order?** Strahler order is a standard hydrological classification system used to define the size and hierarchy of a river network. **Order 1:** Headwater streams (the smallest tributaries at the top of the hillslopes). **Order 2:** Formed when two Order 1 streams join. **Order 3:** Formed when two Order 2 streams join. **Order 4+:** The main river trunk. *(Rule: When a lower-order stream joins a higher-order stream, the higher order remains. It only increases when two streams of the same order meet).*
What is *Si*? It is the Strahler Order of the specific pixel (cell *i*) you are calculating. The model reads your DEM and calculates the stream network. It assigns a value of 1, 2, 3, 4, etc., to every stream pixel. It is just the order of the current pixel being evaluated in the loop.

What is *S*0? It is the Strahler Order at the Reference Catchment River/Stream Outlet (the delta). This is the maximum order of the river at the point where you took your field measurements (e.g., where it enters the Lake). If the main river at the outlet is a 4th-order stream, you set *S*0=4 in the UI.

Why use the ratio (*Si*/*S*0)? The formula does **not** use the raw orders directly. It uses the **ratio** of the local order to the outlet order to scale your field measurements. **At the Outlet:** *Si*=*S*0 (e.g., 4/4=1). The formula gives *Di*=*D*0 (the full measured depth).

Upstream: *Si* is smaller (e.g., a 1st order stream where *S*0=4). The ratio is 1/4=0.251/4=0.25. Raised to the power *f* (e.g., 0.4), the depth scales down dramatically. This perfectly simulates how a tiny headwater stream is much shallower than the massive main river trunk.

Example Calculation: Assume: Outlet order *S*0=4, Outlet depth *D*0=2.0 meters, *f*=0.4.

How does the plugin get these? You do not need to measure them manually. The **River Conductance tab** uses the GRASS r.watershed engine to automatically extract the DEM, route the water, and assign the Strahler order to every pixel based on the flow accumulation. It then applies this formula to scale your *D*0 and *Stage*0 inputs upstream!

Gaining vs. Losing Classification, using the smoothed water table, the model calculates absolute elevations:

- **Streambed Elevation:** $Z_{bed} = DEM_i - D_i$
- **Water Table Elevation:** $Z_{wt} = DEM_i - GWD_i$ *(GWD is smoothed via a 3x3 mean filter to prevent noisy artifacts)*

Classification:

- If $Z_{wt} > Z_{bed}$: Groundwater is above the streambed → **Gaining Stream** (Recharge = 0).
- If $Z_{bed} > Z_{wt}$: Streambed is above the water table → **Losing Stream.**

The model relies on a combination of your Input Rasters, your Field Measurements, and Automatic Calculations:

DEMi (Digital Elevation Model). The elevation of the land surface at pixel *i* (in meters above sea level). Where it comes from: It is read directly from your input raster file located at inputs/maps/dem.tif.

Di (Streambed Depth). The depth of the river channel from the surface down to the riverbed at pixel *i*. Where it comes from: It is calculated by the model using the formula we discussed earlier: $Di=D0\times(Si/S0)^f$. *D*0 comes from your Field Measurements (typed into the UI). *Si* (Strahler order) is automatically derived from the DEM by the GRASS routing engine. *S*0 is the order at the outlet (typed into the UI). *f* is the depth exponent (typed into the UI, default 0.4).

GWDi (Groundwater Depth). The depth from the land surface down to the water table at pixel *i*. Where it comes from: It is read directly from your input raster file located at inputs/maps/gwdepth/gwdepth1.tif (or whichever month is being processed). Original Source: You obtained this from a global dataset (e.g., Fan et al. 2013), a national hydrogeological map, or interpolated well data.

The 3x3 Filter: The model doesn't use the raw raster directly. As the image states, it applies a 3x3 mean filter to smooth out pixel-to-pixel noise (which prevents the model from flip-flopping between "gaining" and "losing" every few meters).

Summary of the Flow: The model looks at the DEM (Surface Elevation). It subtracts your scaled Streambed Depth (*Di*) to find the river bottom. It subtracts the GWD from the DEM to find the Water Table Elevation. It compares the two to decide if the stream is gaining or losing!

Focused Recharge (*Qrech*)

For losing streams, the infiltration rate is calculated using the head difference:

$$Q_{rech} = C_i \times \max(0, (Z_{bed} + Stage_i) - Z_{wt})$$

- **Units:** m^3/day. *(This is converted to* mm/day *over the cell area for the final map).*

Units: *m*3/*day*. *(This is converted to mm/day over the cell area for the final map).*

Detailed Assumptions & Scientific Cautions

For peer-reviewed publication, the following assumptions must be explicitly stated:

- **Steady-State Stage Assumption:** The model assumes a constant reference river stage (*Stage*0) that scales with stream order. It does not dynamically route a flood wave over time.
- **Multi-Channel Deltas:** The model uses *Wout* as the *total effective width*. If your delta has multiple active channels, you must sum their widths when entering *Wout*.
- **Continuous vs. Discrete:** Strahler order is discrete (creating step-jumps at confluences), while drainage area is continuous. We recommend weighting the area exponent (*b*) higher than the order exponent (*a*) to ensure a smooth spatial conductance map.
- **Ephemeral Wadi Limitations:** For ephemeral streams, calculating a steady-state stage is physically artificial. The Stream Typology multiplier (*Ti*) mitigates this by artificially lowering the conductance of ephemeral channels, reflecting their shorter duration of flow.

Step-by-Step Usage Guide

- **Prepare the Boundary:** Ensure your Project Boundary is loaded in the *Inputs* tab.
- **Prepare the Boundary & Geology:** Ensure your Project Boundary and Geology Map (rasterized to the same grid as the DEM) are loaded in the *Inputs* tab. The model will use the geology raster to calculate the Gi*Gi* factor for every pixel.
- **Select Stream Source:**
  - ✓ *Derive from DEM:* Uses the built-in GRASS engine to automatically extract Strahler orders and flow accumulation.
  - ✓ *Vector File:* Upload a shapefile of the river network (Highly Recommended for flat areas).
- **Stream Typology (Optional):** If you have a raster classifying streams as Perennial(1), Intermittent(2), or Ephemeral(3), upload it. This applies a multiplier to the conductance.
- **Enter Field Data:** Input *K*0,*W*0,*L*0,*M*0. **Crucial:** Ensure *L*0 matches your MODFLOW grid cell size (e.g., if grid is 500m, set *L*0=500).
- **Enter Delta Geometry:** Input *Wout* (sum of active delta widths) and *Wexpected* (normal upstream

width).

- **Enter Exponents:**
  - ✓ Leave defaults (a=0.5,b=0.4,c=0.35,d=0.4,f=0.4,g=0.4a=0.5,b=0.4,c=0.35,d=0.4,f=0.4,g=0.4) for a first run.
- Click **"Calculate Conductance & Focused Recharge"**.
- Click **"Combine with Diffusive Recharge"** to generate the final Recharge_Total_Annual.tif for your Groundwater Potential map.

Expected Outputs (Saved in outputs/River_Conductance/)

- ✓ Stream_Conductance.tif: Spatial *Ci* map (*m*2/*day*).
- ✓ Stream_Depth.tif: Streambed incision depth (*m*).
- ✓ Stream_Status_Losing.tif: Binary map (1 = Losing, 0 = Gaining).
- ✓ Stream_Focused_Recharge.tif: Recharge volume (*mm*/*day*).
- ✓ MODFLOW_Stream_Table.csv: The master RIV package table. Contains Layer, Row, Col, Stage_Elev_m, Cond_m2_day, Rbot_Elev_m, and diagnostic columns (Strahler, Area, Slope, Width, Depth, Status, Recharge). Directly importable into ModelMuse.

Parameter Reference Table: Meanings, Units, and How to Obtain Them

To successfully run the River Conductance model, you must populate the UI with physical parameters. The table below explains every variable, its units, and exactly how to obtain or derive it for your catchment.

Symbol	Meaning	Units	How to Obtain (Source)	Typical Range
***K*0**	Reference Riverbed Hydraulic Conductivity	m/day	**Field Measurement:** Permeameter test, slug test, or grain-size analysis of riverbed sediment at the outlet.	10−4 to 102
***W*0**	Reference Normal Channel Width	m	**Field Survey / RS:** Measure the bankfull width perpendicular to the flow at several transects. Use the **median** value.	5 to 100
***L*0**	Reference Reach Length	m	**GIS / Model Grid:** This should ideally match the size of one MODFLOW grid cell (e.g., if MODFLOW cell is 500m, use 500).	100 to 1000
***M*0**	Effective Riverbed Thickness	m	**Field Measurement:** Soil probing, shallow excavation, or geophysics to find the depth of the permeable sediment layer.	0.1 to 5
***D*0**	Reference Streambed Incision Depth	m	**Field Survey / GIS:** The depth of the river channel from the valley floor to the bottom of the streambed at the outlet.	1 to 10

***Stage*0**	Reference River Stage	m	**Field Survey:** The height of the water surface above the streambed during a typical flow event.	0.1 to 5
Wout	Total Delta Outlet Width	m	**Remote Sensing (Google Earth) / Field:** Sum the widths of all active channels where the river enters the delta/lake.	10 to 500
Wexpected	Expected Normal Upstream Width	m	**Remote Sensing (Google Earth):** Measure the single-channel width immediately upstream of the delta expansion.	5 to 100
***A*0**	Outlet Drainage Area	km²	**GIS:** Calculate flow accumulation at the outlet cell × cell area.	101 to 105
***S*0**	Outlet Strahler Stream Order	—	**GIS:** Derived automatically by the plugin's GRASS engine.	1 to 10
***I*0**	Outlet River Slope	m/m	**GIS:** Extract DEM elevations along the outlet reach and divide by the length.	0.0001 to 0.1
Gi	Geological Factor	—	**GIS / Hydrogeology:** Map the underlying geology. *Gi =Kgeology*/*Kreference*. If unknown, use 1.0.	0.1 to 10
Ti	Stream Typology Multiplier	—	**GIS / User Input:** Classify streams as Perennial (1.0), Intermittent (0.8), or Ephemeral (0.5).	0.5 to 1.0
a	Stream-Order Exponent	—	**Literature / Calibration:** Controls how strongly order affects conductance.	0.3 to 0.7 (Default: 0.5)
b	Drainage-Area Exponent	—	**Literature / Calibration:** Controls scaling with catchment size.	0.2 to 0.6 (Default: 0.4)
c	OEI Exponent	—	**Literature / Calibration:** Controls delta influence.	0.2 to 0.5 (Default: 0.35)
d	Slope Exponent	—	**Literature / Calibration:** Controls how flat reaches gain more conductance.	0.2 to 0.6 (Default: 0.4)
f	Depth Exponent	—	**Hydraulic Geometry:** Controls how streambed depth scales.	0.3 to 0.5 (Default: 0.4)
g	Stage Exponent	—	**Hydraulic Geometry:** Controls how stage scales.	0.3 to 0.5 (Default: 0.4)

How to Determine the Exponents (a,b,c,d,f,ga,b,c,d,f,g**):**

The exponents define the "shape" of the river network. They are not universal physical constants, but rather calibration parameters that represent the geomorphology of your specific catchment.

Fast-Track Run:

Use the defaults (a=0.5,b=0.4,c=0.35,d=0.4,f=0.4,g=0.4a=0.5,b=0.4,c=0.35,d=0.4,f=0.4,g=0.4). These are derived from global averages of hydraulic geometry. **How to calibrate:** Run the model with the defaults. Then, run it again with ±50% adjustments (e.g., b=0.2 vs b=0.6). Compare the resulting conductance map to known groundwater levels or baseflow estimates. *If the model underestimates upstream groundwater levels*, **increase** the Area exponent (b) or Stream Order exponent (a) so more water infiltrates upstream. *If the delta is absorbing too much water*, **decrease** the OEI exponent (c).

Important Notes on Units (**To prevent catastrophic scaling errors in MODFLOW):**

Lengths: Ensure all lengths ($W,L,M,D,Stage$) are entered in **meters**.

Areas: Ensure Drainage Area (A_0) is entered in **square kilometers (km^2)**. The code handles the internal conversion.

Conductance Units: The final output (C_i) is in m^2/day, which is the exact unit MODFLOW 6 expects for the RIV package.

Recharge Units: The focused recharge raster is first in mm/day. It is calculated as a daily infiltration rate but is multiplied by the user-defined "Flow Days" to output the final rasters in $mm/year$. This ensures it can be safely added to the Diffusive Recharge without unit conversion errors, when you combine it with the Diffusive Annual Recharge (which is in $mm/year$).

17. References and acknowledgements

17.1 Original model

Abdollahi, K., Bashir, I., & Batelaan, O. (2016). *WetSpass-M: spatially distributed monthly water balance model*. Vrije Universiteit Brussel.

Batelaan, O., & De Smedt, F. (2001). WetSpass: a flexible, GIS-based, distributed recharge methodology for regional groundwater modelling. *IAHS Publication*, 269, 11–18.

Batelaan, O., & Woldeamlak, S. T. (2004). *WetSpass-M documentation*. Vrije Universiteit Brussel.

17.2 Bucket formulation

Arnold, J. G., Srinivasan, R., Muttiah, R. S., & Williams, J. R. (1998). Large-area hydrologic modeling and assessment: Part I. Model development. *Journal of the American Water Resources Association*, 34(1), 73–89. [SWAT soil-water bucket]

Bergström, S. (1992). The HBV model — its structure and applications. *SMHI Reports Hydrology*, No. 4. [HBV soil-moisture routine]

Neitsch, S. L., Arnold, J. G., Kiniry, J. R., & Williams, J. R. (2011). *Soil and Water Assessment Tool theoretical documentation*. Texas Water Resources Institute. [SWAT ET and percolation]

17.3 Evapotranspiration

Monteith, J. L. (1965). Evaporation and environment. *Symposia of the Society for Experimental Biology*, 19, 205–234.

Priestley, C. H. B., & Taylor, R. J. (1972). On the assessment of surface heat flux and evaporation using large-scale parameters. *Monthly Weather Review*, 100(2), 81–92.

17.4 Conductance Related

Leopold, L. B., & Maddock, T. (1953). *The hydraulic geometry of stream channels and some physiographic implications* (Vol. 252). US Geological Survey Professional Paper.

Harbaugh, A. W. (2005). *MODFLOW-2005, the US Geological Survey modular ground-water model: the ground-water flow process*. US Geological Survey.

Langevin, C. D., et al. (2017). *Documentation for the MODFLOW 6 Groundwater Flow Model*. US Geological Survey.

O'Brien, N., et al. (2021). *River-aquifer exchange conductance: A review of estimation methods and*

their application to regional-scale models. Journal of Hydrology.

17.5 Acknowledgements

The modernization and QGIS plugin effort was carried out by Melese. G. Gurmu (2026, Addis Ababa, Ethiopia). The original physics remain the intellectual property of K. Abdollahi, I. Bashir, and O. Batelaan, to whom co-authorship should be given in any publication that uses Wetspass-M or Wetspass-M Modern. DeepSeek AI is used for programming the current scripts, with a careful guidance. **AI Programming Assistant:** DeepSeek AI (used for programming the current scripts under careful Author's guidance).

Appendix A — Glossary

AET (Actual Evapotranspiration) — The sum of transpiration, soil evaporation, open-water evaporation, impervious-surface evaporation, and (in shallow groundwater settings) groundwater evaporation over a period.

Addition (Combine) — The default method in v2.4.1 for calculating Total Recharge: R_total = R_diffusive + R_focused. The two layers are geometrically aligned, and the focused recharge is added to the diffuse recharge to account for wadi transmission losses.

Auto-Cropping — A memory-management feature that physically clips all input rasters to the bounding box of the project boundary shapefile *before* any mathematical calculations begin. This prevents MemoryError crashes on large national or regional datasets.

Bucket (Soil Bucket) — A simplified representation of the soil column as a single reservoir of plant-available water. It tracks soil moisture from month to month.

Conductance (C*C*) — A measure of how easily water can flow between a river and the underlying aquifer (units: m2/day*m*2/*day*). It is calculated based on riverbed hydraulic conductivity, riverbed thickness, channel width, and reach length.

Closure error (Mass Balance Closure) — The residual in the water balance equation after all fluxes and storage changes are accounted for (P−AET−SR−R−ΔS*P*−*AET*−*SR*−*R*−Δ*S*). Should be close to zero on a well-posed model run (typically < 2 mm/yr).

DEM (Digital Elevation Model) — A raster of land-surface elevations. Used to derive slopes, flow direction, stream networks, and valley floors.

Flow Days — A user-defined parameter in the River Conductance module representing the number of days per year that a stream actually flows. Used to convert daily infiltration rates into annual recharge volumes (mm/year*mm*/*year*).

Focused Recharge — Recharge concentrated on a small fraction of the landscape, typically streambeds and topographic lows (wadis), as opposed to diffuse recharge spread over the whole catchment. Calculated using the conductance formula and the head difference between the river stage and the water table.

Gaining Stream — A stream that receives water from the aquifer along its reach. Classified in the model when the absolute Water Table Elevation is *higher* than the Streambed Elevation (Zwt>Zbed*Zwt*>*Zbed*). Focused recharge is zero for gaining streams.

Geological Factor (Gi*Gi*) — A spatial multiplier applied to riverbed conductance based on the permeability of the underlying rock or sediment (e.g., clean alluvium = 2.0, massive granite = 0.1). Provided as a raster aligned to the DEM.

GWD (Groundwater Depth) — The vertical distance from the land surface down to the water table. It is

provided as a monthly input raster.

Losing Stream — A stream that loses water to the aquifer along its reach. Classified in the model when the absolute Streambed Elevation is *higher* than the Water Table Elevation (Zbed>Zwt*Zbed*>*Zwt*). Losing reaches are the source of focused recharge.

MFD (Multi-Flow Direction) — A hydrological routing algorithm where water from a pixel splits and flows into multiple downstream neighbors based on slope. It is used in the plugin to resolve flat-area routing artifacts and ensure realistic, meandering stream networks (as opposed to single-flow D8).

OEI (Outlet Expansion Index) — A multiplier that accounts for the physical widening of a river as it enters a delta or lake. It is calculated as Wout/Wexpected*Wout*/*Wexpected*. It prevents the model from artificially over-estimating upstream conductance.

PET (Potential Evapotranspiration) — The evapotranspiration that would occur if water were freely available. Used with Precipitation to calculate the Aridity Index.

Spin-Up — The process of running a state-tracking model repeatedly on the same inputs until the internal soil moisture state reaches a repeating annual cycle. The workflow defaults to a maximum of 30 cycles but exits early automatically once convergence is reached (ΔSΔ*S* < tolerance).

Stream Typology — A classification of streams based on flow duration (Perennial = 1.0, Intermittent = 0.8, Ephemeral = 0.5). It applies a multiplier to both the river stage and conductance.

Streambed Depth (Di*Di*) — The physical incision depth of the river channel below the valley floor. It is calculated dynamically using the DEM and field-measured reference depth (D0*D*0) scaled by Strahler order.

UNEP 5-Zone Climate Capping — A physics safeguard that overrides raw groundwater depth inputs based on the UNEP Aridity Index (P/PET). It forces deep water tables in arid regions to prevent infinite, non-physical groundwater evaporation and subsequent mass balance failures. Can be toggled off for humid or lake-adjacent catchments.

ΔSΔ*S* (Change in Soil Storage) — The change in soil-water storage over a period. Zero in quasi-steady-state models; non-zero in dynamic models (like this one). Accounted for in the mass balance equation.

Appendix B — File listing

Folder / Filename	Purpose
Root Directory	
__init__.py	QGIS plugin entry point (contains version and author metadata).
metadata.txt	Plugin metadata (name, version, changelog) read by the QGIS Plugin Manager.
icon.png, icon.ico	Icons used in the QGIS toolbar and plugin menu.
LICENSE.txt	Academic-use license.
README.md	Project overview, requirements, and quick-start guide.
CHANGELOG.md	Version history and release notes.
wetspass_plugin.py	Main plugin class (initializes the toolbar button and menu).
wetspass_provider.py	QGIS Processing provider (integrates the tool into the QGIS toolbox).
wetspass_algorithm.py	QGIS Processing algorithm (allows batch/scripted runs).
core/	
calculation_engine.py	Core physics (AET, interception, runoff, and soil bucket math).
workflow.py	The orchestrator. Manages spin-up cycles, checkpoints, resume logic, auto-cropping, and MODFLOW output aggregation.
raster_io.py	Raster reading/writing with automatic block-processing to prevent memory crashes.
cropper.py	Boundary cropping engine used for the final outputs.
gw_depth_handler.py	**NEW:** Handles the UNEP 5-Zone Climate Capping and the Smart Cache (auto-clears when settings change).
river_conductance.py	**NEW:** The field-anchored geomorphic scaling engine. Calculates conductance, stream depth/stage, and generates the MODFLOW CSV.
focused_recharge.py	Focused recharge engine (routes runoff into streambeds to calculate transmission losses).
stream_derivation.py	Hydrological routing. Upgraded to **Multi-Flow Direction (MFD)** to solve flat-area straight-line artifacts.
stream_io.py	Rasterizes vector stream shapefiles onto the DEM grid.
run_report.py	Writers for the RunSummary.txt and _run_metadata.json files.
ui/	
main_dialog.py	The main GUI. Contains all 14 tabs, the Green Run button, the math formula box, and progress bars.
worker.py	Background thread for the Diffusive model (includes the auto-cropping logic).
focused_tab.py	The Focused Recharge UI tab.
focused_worker.py	Background thread for the Focused model.
water_balance_tab.py	The Water Balance panel (calculates and visualizes the mass balance closure).
calibration_tab.py	Calibration panel (for comparing simulated vs. observed data).
boundary_preview.py	Widget that draws a preview of the project boundary shapefile.
tools_menu.py	Actions for the tools at the bottom of the UI (e.g., Open Report, Crop outputs).
help/	
WetSpassM_UserGuide.pdf	This document.

Appendix C - Detailed Breakdown of Each Coefficient (River Bed Conductance)

a — *Strahler Stream-Order Exponent*

- ✓ **Indicator:** Controls the **discrete step-jump** in conductance at river confluences (e.g., when two Order-2 streams become an Order-3 stream).
- ✓ **Fast-Track Default:** 0.5
- ✓ **Recommended Range:** 0.3 to 0.7
- ✓ **How to select:** If your stream gauges show that river width doubles exactly at confluences, use a higher a. However, because Strahler order creates abrupt spatial jumps, it is generally recommended to keep a lower (e.g., 0.3) and rely more heavily on the continuous Area exponent (b) for a smoother map.

b — *Drainage-Area Exponent*

- ✓ **Indicator:** Controls the **continuous scaling** of conductance as the catchment area grows. Larger upstream contributing areas provide more water, thus increasing conductance.
- ✓ **Fast-Track Default:** 0.4
- ✓ **Recommended Range:** 0.2 to 0.6
- ✓ **How to select:** This is usually the most important parameter. If your model underestimates upstream groundwater heads, increase b (e.g., 0.5 or 0.6). If the upstream tributaries are receiving too much focused recharge, lower b.

c — *Outlet Expansion (OEI) Exponent*

- ✓ **Indicator:** Controls how strongly the measured delta expansion at a Lake affects the reference conductance at the outlet.
- ✓ **Fast-Track Default:** 0.35
- ✓ **Recommended Range:** 0.2 to 0.5
- ✓ **How to select:** If your MODFLOW model shows that the delta is absorbing too much water (or if the lake levels are dropping), decrease c to 0.2. If the delta is drying out too fast, increase c to 0.45.

d — *Longitudinal Slope Exponent*

- ✓ **Indicator:** Controls the **inverse relationship** between slope and conductance. The formula uses $(I_0/I_i)^d$. Because it is inverted, *flatter* reaches get *higher* conductance (water lingers longer, allowing more infiltration).
- ✓ **Fast-Track Default:** 0.4
- ✓ **Recommended Range:** 0.2 to 0.6
- ✓ **How to select:** If you have field data showing that your steep mountain streams lose just as much water as your flat valley streams, set d to 0.0 (which disables the slope effect). If you want the flat valley to dominate the recharge, increase d to 0.5 or 0.6.

f — ***Streambed Depth Exponent***

- ✓ **Indicator:** Controls how quickly the riverbed incision depth decreases as you move upstream to smaller tributaries. The exponent f controls how aggressively this incision depth decreases as you move from the large outlet to the small headwaters.
- ✓ **Fast-Track Default:** 0.4
- ✓ **Recommended Range:** 0.3 to 0.5
- ✓ **How to select:** Use your field data. If you know the main river is 3m deep, but a 1st-order stream is only 0.5m deep, a f value of 0.4 will replicate this physical reality.
- ✓ **If f is low (e.g., 0.2):** The upstream tributaries remain relatively deep. The model assumes the incision depth is almost uniform across the whole catchment.
- ✓ **If f is high (e.g., 0.8):** The upstream tributaries become extremely shallow very quickly. The model assumes only the main river is deeply incised.

How to Select f (The Mathematical Method)

While you can guess f based on experience, the most scientifically defensible way to select it is to use your field data and solve for f mathematically:

$$f = \frac{\log(D_{tributary}/D_{outlet})}{\log(S_{tributary}/S_{outlet})}$$

Example Calculation **(Correcting guide's example):** Let's say you measured:

- ✓ **Outlet ($S0=4$ $S_0=4$):** Depth $D0=3.0$ $D_0=3.0$ meters.
- ✓ **Tributary ($Si=1$ $S_i=1$):** Depth $Di=0.5$ $D_i=0.5$ meters.

If you plug these into the formula to find the *exact* f that replicates your field data:

$$f = \frac{\log(0.5/3.0)}{\log(1/4)} = \frac{\log(0.1667)}{\log(0.25)} = \frac{-1.79}{-1.38} \approx \mathbf{1.30}$$

(Note: If you use the fast-track default of $f=0.4$, the model would calculate the 1st-order tributary depth as $3.0\times(0.25)^{0.4}=\mathbf{1.72}$ meters. So $f=0.4$ assumes the upstream tributaries are much deeper than your field data suggests).

g — ***River Stage Exponent***

- ✓ The formula $Stage_i=Stage_0\times(S_i/S_0)^g$ uses g to control how aggressively the water depth drops as you move upstream.
- ✓ **Indicator:** Controls how the water depth (stage) scales with stream order.
- ✓ **Fast-Track Default:** 0.4

- ✓ **Recommended Range:** 0.3 to 0.5
- ✓ **How to select:** Similar to depth. If your ephemeral streams have very shallow flash floods compared to the main river, keep g around 0.3. **If $g=0.3$ (Low):** The stage remains relatively similar across the whole network. This assumes that a flash flood fills both the main river and the headwater wadis to roughly the same depth. **If $g=0.7$ (High):** The stage drops off very rapidly. This assumes the main river is deep (e.g., 3 meters), but the 1st-order headwaters only have 5 centimeters of water.

Appendix D - Indicators for Calibration (What to look for)

When you run the model with your chosen coefficients, use these indicators to know if you need to adjust them:

- ✓ **Upstream Groundwater Heads (MODFLOW):** If your MODFLOW model shows groundwater heads dropping drastically in the headwaters, your upstream conductance is too low. **Increase b and a**.
- ✓ **River Leakage (MODFLOW):** If the MODFLOW RIV package shows that the river is losing 100% of its flow within the first 5 km (drying up too fast), your conductance is too high. **Decrease b and c**.
- ✓ **Baseflow:** If your observed baseflow in the dry season is higher than what the model simulates, your conductance might be too low, meaning the aquifer isn't receiving enough focused recharge to sustain the stream. **Increase the Area exponent b**.
- ✓ **Delta / Lake Interaction:** If Lake under consideration is receiving too much groundwater discharge in your model, **decrease the OEI exponent c**.

****